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11 June 1971
SUPERSEDING
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(See 6.7)

MILITARY SPECIFICATION

PLASTIC SHEETS, LAMINATED THERMOSETTING, NYLON FABRIC BASE, PHENOLIC-RESIN

This specification is mandatory for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers type NPG plastic laminated thermosetting sheet material intended primarily for electrical insulating purposes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
- MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY
- MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL
- FED-STD-406 - Plastics: Methods of Testing.

MILITARY
- MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
- MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
- D 150 - A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials, Tests for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Construction. The material shall consist of nylon cloth filler properly impregnated with phenolic resin compound or binder, processed to conform to this specification.

3.1.2 Machinability. The material shall be such that it can be drilled, tapped, sawed, and machined in all directions in accordance with the manufacturer's recommended technique without cracking, splitting, or otherwise impairing the material for general use.

3.1.3 Color. The material shall be natural in color. (Natural is the color produced by the natural, undyed base or filler and the resin used.)

3.1.4 Surface defects. The material shall be free from blisters, wrinkles, or cracks, and reasonably free from other small defects such as scratches, dents, and heat marks.

3.1.5 Property values. The material shall conform to the property value specified in table IV (see 4.3.1). The values obtained for each set of specimens taken from the same sheet shall be averaged before comparison with the applicable table.

3.1.6 Uniformity. All sheets of corresponding thickness shall be uniform in texture, finish, and specified properties.

3.1.7 Surface finish. Unless otherwise specified (see 6.2), the surface finish of sheets shall be semigloss as produced by the laminating operation.

3.1.8 Dimensions and tolerances.

3.1.8.1 Thickness. Sheets shall be furnished in the nominal thickness shown in table I, as specified (see 6.2). At least 90 percent of the area of the sheet shall be within the variations specified in table I, and at no point shall the thickness as measured vary from the nominal thickness by a value greater than 125 percent of tolerance specified in table I.

Table I - Thickness.

Nominal thickness 1/		Permissible variations (inch) (±)
Inch	Decimal equivalent	
0.010	0.010	0.003
.015	.015	.0035
.020	.020	.004
.025	.025	.0045
1/32	.031	.0065
3/64	.047	.0075
1/16	.0625	.0075
3/32	.094	.009
1/8	.125	.010
5/32	.156	.011
3/16	.1875	.0125
7/32	.219	.014
1/4	.250	.015
5/16	.3125	.035
3/8	.375	.040
7/16	.438	.044
1/2	.500	.048
5/8	.625	.054
3/4	.750	.058
7/8	.875	.062
1	1.000	.066

1/ On sheets of nominal thickness not listed in table I, the permissible variations shall be the same as for the next greater thickness.

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3.1.8.2 Length and width. Unless otherwise specified in the contract or order, the manufacturer's standard sizes between 36 and 50 inches in width and between 36 and 96 inches in length will be acceptable. The length and width of sheets may vary 1 inch over or under the manufacturer's standard size. Non-uniform standard sheet dimensions caused by cutting specimens for test required by this specification shall not be cause for rejection, unless particular dimensions are specified. When particular sheet dimensions are specified (see 6.2), the permissible variations from the specified length or width shall be as specified in table II.

Table II - Permissible variations in length or width.

Nominal thickness	Permissible variations in length or width (+)		
	6 inches and under	Over 6 to under 24 inches	24 inches and over
Inch	Inch	Inch	Inch
Up to 1/4, inclusive	0.010	0.015	1/32
17/64 to 1/2, inclusive	.012	.017	1/32
33/64 to 1, inclusive	.015	.020	1/32

3.1.9 Warp or twist. (Applicable only to sheets 36 inches and greater in length and 36 inches and greater in width.) The warp or twist of material as delivered, shall not exceed that specified in table III (see 4.5.9). Percentage of warp is specified in terms of the lateral dimensions (length and width) of the material, percentage of twist is specified in terms of dimensions from one corner to the opposite corner.

Table III - Warp or twist.

Thickness	Permissible variation, on basis of 36 inch dimension
Inch	Maximum percent
1/32 to under 1/16	5.00
1/16 to under 1/8	2.50
1/8 to 1/4, inclusive	1.00
Over 1/4, up to and including 3/4	0.50
Over 3/4	0.25

3.2 Marking. Each full size sheet shall be legibly marked with the manufacturer's name or trade-mark, and the type. The method of marking shall be satisfactory to the procuring activity.

3.3 Workmanship. The sheets shall be manufactured and processed in a careful and workmanlike manner, and shall be free from defects which may affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Each sheet of every lot shall be subjected to surface inspection to determine compliance with 3.1.4. If any of the sheets fail to conform to this requirement, such sheets shall be rejected.

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4.3 Quality conformance testing.

4.3.1 Quality conformance tests shall be conducted on material selected from each production lot. A production lot shall consist of all sheets of a given machine run and of a particular thickness range as shown in table IV and offered for inspection at one time. A machine run shall consist of all the material pressed from a coating operation in which the basic resin and filler treating conditions are the same. Sampling for tests will be in accordance with level S-1 of MIL-STD-105. Test specimens will be cut from each sample unit for testing in accordance with the applicable characteristics of table IV to determine conformance of the sample with the requirements of this specification.

4.3.2 Acceptance or rejection of lots. If any of the sample sheets tested fail any of the tests shown in table IV, the entire lot shall be rejected. At the option of the manufacturer a rejected lot may be resubmitted for lot acceptance inspection and tests, but only after the manufacturer has subjected specimens from each sheet in the lot to the lot acceptance test wherein failure was observed and has removed all nonconforming sheets.

4.3.3 Reports of quality conformance tests. Acceptance or rejection of lots will normally be determined by the inspector on the basis of the lot acceptance test report, certified by the inspector, and shall be forwarded to the procuring activity with a statement as to whether the lot was accepted or rejected.

4.4 Conditioning of test specimens.

4.4.1 Nomenclature. The following letters shall be used to indicate the respective general conditioning procedures:

Condition A - As received; no special conditioning.
Condition C - Humidity conditioning.
Condition D - Immersion conditioning in distilled water.
Condition E - Temperature conditioning.

Whenever a conditioning letter is followed by an inferior 1, as D_1, a prior temperature conditioning has been carried out.

4.4.2 Designation. Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general condition of the specimen, that is, as received, humidity, immersion, or temperature conditioning.
(b) A number indicating in hours the duration of the conditioning.
(c) A number indicating in degrees centigrade the conditioning temperature.
(d) A number indicating relative humidity whenever relative humidity is controlled.

The numbers shall be separated from each other by a slant mark, and from the capital letter by a dash.

Examples:
Condition C-96/35/90 - Humidity condition, 96 hours at 35°C. and 90 percent relative humidity.
Condition D-48/50 - Immersion condition, 48 hours in distilled water at 50°C.
Condition E-48/50 - Temperature condition, 48 hours at 50°C.

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Table IV - Quality conformance tests.

Property to be tested	Test paragraph	Number of specimens for each sample	Tests per specimen	Conditioning (see 4.4)	Unit of value	Values required for each sample thickness range[5] (inches)					
						0.031 or less	0.032 to 0.094	0.095 to 0.125	0.126 to 0.250	0.251 to 0.499	0.500 to 1.000
Dielectric breakdown parallel to laminations, step-by-step test.	4.5.2	1[1]	1	D-48/50	Minimum kilovolts ---	40.0	40.0	40.0	40.0	40.0	40.0
		4[2]	1								
Flexural strength, tested flatwise----	4.5.4	--	--	A							
Cut lengthwise ---	---	4	1	----	Minimum pounds per square inch.	10,000[3]	10,000	10,000	9,000	9,000	9,000
Cut crosswise ----	---	4	1	---		9,500[3]	9,500	9,500	8,500	8,500	8,500
Bonding strength ---	4.5.5	4	1	A	Minimum pounds -----	[4]	[4]	[4]	[4]	[4]	1,000

[1] Short-time.

[2] Step-by-step.

[3] Not required for sizes under 0.032 inch in thickness.

[4] Not required.

[5] These ranges are for nominal thicknesses, subject to the tolerances specified in table I.

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4.4.3 Time tolerances. Oven conditioning shall be followed by a cooling to room temperature (23°C.) in a desiccator and immersion conditioning shall be followed by cooling to room temperature in distilled water, as specified in table V.

Table V - Conditioning time tolerances.

Conditioning		Cooling		Comments
Condition	Time tolerance	Time	Time tolerance	
E-48/50	-0 hours +2 hours	16 hours or more.	-0 hours	Cool in desiccator. Start test within 1/2 hour after removing specimen from desiccator.
D-48/50 D-24/23	-0 hours +1/2 hour	1 hour	-0 hours +2 hours	Cool by immersion in a sufficient quantity of distilled water to reduce the temperature to 23°C. within 1 hour. Remove individually as needed, and wipe surface water off with a cloth. Start test within 1 minute after removing specimen from water.
E-1/105	-0 minutes +6 minutes	2 hours or more.	-0 hours	See water-absorption test. (See 4.5.6)
D_1-24/23	-0 hours +2 hours	------	---------	See water-absorption test. (See 4.5.6)
C-96/35/90	-0 hours +2 hours	------	---------	Tests after humidity conditioning shall be made on specimens in the humidity chamber. (See 4.5.8) Forced air circulation shall be used in the humidity chamber.

4.4.4 Temperature tolerances. The conditioning temperature tolerances shall be as follows:

Nominal temperature, °C.	Tolerance (±) °C.
23	2
35	1
50	2
105	2

4.4.5 Humidity tolerances. Tolerance on the nominal relative humidity of 90 percent shall be plus or minus 2 percent.

4.5 Test methods.

4.5.1 Definitions. In conducting tests, the term "lengthwise" shall be interpreted to mean that sheet direction known to be the stronger in flexure. "Crosswise" shall then be the sheet direction known to be the weaker in flexure.

4.5.2 Dielectric breakdown parallel to laminations (step-by-step).

4.5.2.1 Specimens. The test specimens shall be 2 by 3 inches by the thickness of the sheet. American Standard tapered pins having a taper of 1/4 inch per foot shall be used. For test specimens having a thickness up to 1/2 inch, inclusive, No. 3 American Standard tapered pins[1/] 3 inches long and having a diameter at the large end of 7/32 inch shall be used. Drill 3/16 inch diameter holes, centrally located, 1 inch apart, center to center, and perpendicular to the faces of the specimen. Using standard tapered pin reamer, ream holes to a sufficient depth to allow the pins to extend approximately 1-1/2 inches from the small end of the hole. For test specimens having a thickness over 1/2 up to 1 inch, inclusive, No. 4 American Standard tapered pins 4 inches long and having a diameter at the large end of 1/4 inch shall be used. Drill two 3/16 inch diameter holes, centrally located,

1/ For information on tapered pins, see Kents Mechanical Engineers Handbook, 12th Edition, Design and Production, page 15-14. (This handbook may be obtained from John Wiley and Sons, Inc., 440 - 4th Avenue, New York, N.Y. 10016.)

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1 inch apart, center to center, and perpendicular to the faces of the specimen. Using standard tapered pin reamer, ream holes to a sufficient depth to allow the pins to extend approximately 1 inch from the small end of the hole. The electrodes shall be inserted after the conditioning of the test specimen. Spheres having a 1/2 inch diameter, when placed on the extremities of the tapered pins, will decrease the tendency to flashover.

4.5.2.2 Procedure. All tests shall be made under oil, at commercial power frequency of 60 cycles. One specimen only shall be tested by the short-time method for the purpose of providing a basis for the initial voltage applied in the step-by-step test. In the short-time test, starting at zero, the testing voltage shall be increased as uniformly as possible at a rate of 500 volts per second. On the remaining specimens, the step-by-step method shall be used. The initial voltage applied in the step-by-step test shall be 50 percent of the short-time breakdown voltage. The initial voltage in the step-by-step tests, shall be applied for 1 minute and the voltage shall then be increased in increments as follows, holding the voltage at each step for 1 minute:

Breakdown voltage by short-time method (kilovolts)	Increment of increase for step-by-step method (kilovolts)
12.5 or less	0.5
12.5 to 25	1.0
25 to 50	2.5
50 to 100	5.0

4.5.2.3 Report. The breakdown voltage shall be reported in kilovolts and the time required to break each specimen shall be reported in seconds.

4.5.3 Impact strength (tested edgewise).

4.5.3.1 Specimens. Specimens of sheets in thicknesses of 1/8 to 1/2 inch, inclusive, shall be tested in sizes 2-1/2 inches by 1/2 inch by thickness without build-up. Each specimen shall be notched in one edge (side). The notch shall be centrally located with respect to the ends of the specimen. The angle of the notch shall be 0.400 ± 0.005 inch with a curvature at the bottom of the notch of 0.010 ± 0.002 inch. Different sets of specimens shall be cut lengthwise and crosswise of the sheet.

4.5.3.2 Procedure. After the conditioning of the specimen, the impact strength shall be determined on a pendulum type Izod impact machine. Accurate corrections shall be made for friction and windage losses. The specimen shall be located in the machine by means of a jig, with center line of notch on the level of the top of the clamping surface. In the striking position, the striking edge of the pendulum shall be 0.866 inch from the edge of the specimen clamp. The pendulum shall be released from such a position that the linear velocity of the striking edge at the instant of impact shall be approximately 11 feet per second, which is the linear velocity corresponding to an initial elevation of the striking edge of 2 feet. The blow shall be struck on the notched side. The pointer setting shall be recorded and corresponding correction shall be obtained from a chart. This chart shall be constructed by measuring the friction and windage losses, using the proportionate amount of these losses as the correction factor.

4.5.3.3 Calculations. The impact strength shall be taken as the energy absorbed in breaking the specimen. It shall equal the difference between the energy remaining after breaking the specimen and the pendulum's initial energy, and shall be expressed in foot-pounds per inch of notch.

4.5.4 Flexural strength (tested flatwise).

4.5.4.1 Specimens. Specimens of sheets in thicknesses of 1/32 to 2 inches, inclusive, shall be tested flatwise in accordance with method 1031 of FED-STD-406, except that the dimensions and speed of test specified in table VI shall be used. Different sets of specimens shall be cut lengthwise and crosswise of the sheet.

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Table VI - Dimensions of specimen and speed of test.

Nominal specimen thickness	Width of specimen	Length of specimen	Span[1/]	Speed of test per minute
Inch	Inch	Inches	Inches	Inch
1/32	1	2-1/2	5/8	0.025
1/16	1	3	1	.026
3/32	1	3-1/2	1-1/2	.040
1/8	1	4	2	.053
3/16	1/2	5	3	.080
1/4	1/2	6	4	.106
3/8	1/2	8	6	.160
1/2	1/2	10	8	.213
3/4	3/4	14	12	.320
1	1	18	16	.426

[1/] This span depth ratio is greater than 16 to 1 in order to give clearance between moving head and specimen supports.

4.5.4.2 Procedure. The breadth and depth of the specimens shall be measured to the nearest 0.001 inch. The span length shall be measured to the nearest 0.01 inch. No modules data need be taken. Care shall be taken that the specimens are centered properly in the jig prior to being tested.

4.5.4.3 Calculations. The maximum fiber stress shall be calculated as follows:

$$\text{Maximum fiber stress} = \frac{3PL}{2bd^2}$$

where: P = breaking load in pounds.
L = the span in inches.
b = breadth of specimen in inches.
d = depth of specimen in inches.

4.5.5 Bonding strength.

4.5.5.1 Specimens. Specimens shall be 1 by 1 by 1/2 inch in size. Specimens from sheets over 1/2 inch thick shall be formed by machining down both surfaces.

4.5.5.2 Procedure. The thickness of the specimens shall be recorded. The test machine shall be fitted with a head containing a steel ball 10 millimeters in diameter. After the specimen has been conditioned, it shall be placed on edge and centered accurately beneath the steel ball so that the load is applied in the lengthwise direction of the specimen. The load shall be applied to the specimen until failure occurs. The speed of the test shall be 0.050 inch per minute.

4.5.5.3 Report. The bonding strength shall be reported in pounds.

4.5.6 Water absorption.

4.5.6.1 Specimens. The specimens shall be 3 inches by 1 inch by thickness.

4.5.6.2 Procedure. The specimens shall be weighed individually and then subjected to 105°C. for 1 hour, cooled in a desiccator for at least 2 hours, and reweighed (W_1). The specimens shall then be immersed in distilled water for 24 hours at 23°C. They shall be removed individually as needed, all surfaces wiped off with a dry cloth, and then reweighed immediately (W_2). The test shall begin not more than 1 minute after removal of the specimens from the water and shall be completed as rapidly as consistent with accuracy. Specimens 1/16 inch or less in thickness shall be placed in a weighing bottle immediately after wiping and shall be weighed in the bottle. No correction shall be made for water-soluble matter.

4.5.6.3 Calculations. The percentage of water absorption shall be calculated as follows:

$$\text{Percentage of water absorption in 24 hours} = \frac{W_2 - W_1}{W_1} \times 100$$

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4.5.7 Dielectric constant and dissipation factor (at 1 megacycle).

4.5.7.1 Specimens. The specimens for dielectric constant and dissipation factor tests at 1 megacycle frequency, perpendicular to laminations, shall be 4 inch diameter disks or 4 inch squares cut from sheets 1/8 inch in thickness.

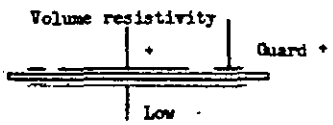
4.5.7.2 Procedure. After the conditioning of the specimens, the electrical measurements shall be made by using a suitable bridge method, a twin-T impedance-measuring circuits, or a resonant-circuit substitution method. The method used shall have an accuracy of determination of dielectric constant of plus or minus 5 percent and an accuracy of determination of dissipation factor of plus or minus 5 percent but in no case closer than 0.001. The specimens shall have been covered on both sides with metal foil using a thin layer of petrolatum as an adhesive or with conductive silver paint electrodes. The dissipation factor shall be directly computed from the instrument readings and shall be considered nominally equivalent to the power factor.2/

4.5.8 Volume resistivity and surface resistance.

4.5.8.1 Specimens. The specimens for volume-resistivity and surface-resistance tests shall be 4 inches by 4 inches by thickness.

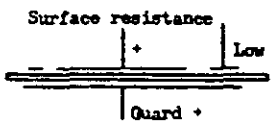
4.5.8.2 Procedure.

4.5.8.2.1 Volume resistivity. The test electrodes shall be made of silver paint sprayed or brushed on the surfaces of the specimen. The upper electrode shall be in the shape of a circle 2 inches in diameter. A guard ring 1/4 inch wide shall be located concentrically with respect to the upper electrode and shall be spaced 1/4 inch from it. The bottom electrode shall be in the shape of a circle 3 inches in diameter. The centers of the upper and lower electrodes shall be centered as accurately as possible so that they are on the same axis. The potentials applied shall be as follows:



The volume resistance shall be measured by means of a 500 volt megohm bridge exactly 1 minute after the current is applied. Tests following the humidity conditioning shall be made on specimens in the humidity chamber.

4.5.8.2.2 Surface resistance. Surface resistance measurements shall be made on the same specimens as those used for volume resistivity. The test procedure shall be similar to that specified for volume resistivity, except that the potentials of the guard and bottom electrodes shall be interchanged as follows:



The surface resistance shall be measured by means of a 500 volt megohm bridge exactly 1 minute after the current is applied. Tests following the humidity conditioning shall be made on specimens in the humidity chamber.

2/ For a complete discussion of theory, apparatus, electrodes, and calculation, refer to American Society for Testing and Materials test method D 150 - A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials, Tests for.

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4.5.8.3 Calculations. The volume resistivity shall be calculated as follows:

$$r = \frac{RA}{L}$$

where: r = volume resistivity in megohm centimeters.
R = measured volume resistance in megohms.
A = area of guarded electrode in square centimeters.
L = average thickness of specimen in centimeters.

Surface resistance shall be reported in megohms.

4.5.9 Warp and twist. The warp and twist test shall conform to method 6051 of FED-STD-406 (see 3.1.9).

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, laminated sheets shall be wrapped individually or interleaved to protect them from abrasion, using not less than 25-pound basis weight kraft paper.

5.1.2 Level C. Packaging of plastic sheets shall be sufficient to afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The supplier's commercial packaging procedures may be utilized when such meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified in the contract or order, the laminated sheets, packaged as specified in 5.1, shall be packed in cleated plywood boxes or nailed wood boxes conforming to PPP-B-601 or PPP-B-621, respectively. Corrugated fiberboard pads and liners shall be used to protect the exposed surfaces and edges of the sheets. Shipping containers shall be lined with a case liner fabricated from material conforming to MIL-L-10547. Seams and closures shall be made with water resistant adhesive conforming to MMM-A-260 and shall have strength and water resistance equal to that of the body material. The gross weight shall not exceed 150 pounds unless the weight of a single sheet exceeds this limit, in which case only one sheet shall be packed in a container.

5.2.2 Level B. Unless otherwise specified in the contract or order, the laminated sheets, packaged as specified in 5.1, shall be packed in a wood-cleated solid fiberboard box conforming to PPP-B-591, or in nailed wood or cleated plywood boxes conforming to PPP-B-621 or PPP-B-601, respectively. Corrugated fiberboard pads and liners shall be used for interior packing. The gross weight shall not exceed 150 pounds unless the weight of a single sheet exceeds this limit, in which case only one sheet shall be packed in a container.

5.2.3 Level C. Containers shall be of uniform size and shall contain, as nearly as possible, the identical number of sheets of one size and type. Containers shall be designed to fit the contents in a compact manner.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Plastic material covered by this specification is intended for use in electronic applications requiring moisture resistance but not heat, flame, and arc resistance. This material shows severe decrease in flexural strength at temperatures above 75°C., and, consequently, must be used with caution at such temperatures.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Surface finish required (see 3.1.7).
(c) Sheet size and thickness required (see 3.1.8.1).
(d) Whether sheets, smaller in dimension than specified in the procurement document due to cutting specimens for quality conformance tests are acceptable (see 3.1.8.2).
(e) Levels of packaging and packing required (see 5.1 and 5.2).
(f) Special marking, if required (see 5.3).

6.3 Fabricated parts or equipment. Requirements applicable to fabricated parts or equipment incorporating nylon-cloth, phenolic-resin, laminated material should be specified in separate specifications or as part of contracts or orders.

6.4 Engineering information. Table VII gives engineering information for properties of the material covered by this specification. This information will not be used for approval action under this specification nor for acceptance or rejection of lots under this specification.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-STD-17C. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "‡" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Custodians:
Army - EL
Navy - SH
Air Force - 84
Review activities:
Army - EL, MR, MU, MI
Navy - SH, EC
Air Force - 84
User activities:
Navy - MC

International Standardization (see 6.6)

Preparing activity:
Navy - SH
(Project 9330-0482)

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Table VII - Engineering information for properties of type NPG laminated material.

Property	Condition	Unit of value	Average value or range	Test procedure
Specific gravity	A		1.14-1.18	Method 5011 of FED-STD-406
Tensile strength:				
Lengthwise direction	A	Pounds per square inch.	5,420-12,000	Method 1011 of FED-STD-406
Cross direction	A	Pounds per square inch.	3,880-14,100	Method 1011 of FED-STD-406
Compressive strength:				
Flatwise	A	Pounds per square inch.	16,800-25,000	Method 1021 of FED-STD-406
Edgewise				
Flexural strength, flatwise:				
Lengthwise direction	A	Pounds per square inch.	9,140-32,400	Method 1031 of FED-STD-406
Cross direction	A	Pounds per square inch.	8,500-19,700	Method 1031 of FED-STD-406
Izod impact strength:				
Flatwise	E-48/50	Foot-pounds per inch notch.	5.94-11.40	Method 1071 of FED-STD-406
Edgewise	E-48/50	Foot-pounds per inch notch.	2.0-8.09	Method 1071 of FED-STD-406
Hardness, Rockwell M	A		84.4-113	Method 1081 of FED-STD-406
Water absorption:				
1/16-inch thickness	E-1/105, followed by D_1-24/23.	Percent	0.220-0.57	4.5.6
1/8-inch thickness	E-1/105, followed by D_1-24/23.	Percent	0.136-0.400	4.5.6
1/2-inch thickness	E-1/105, followed by D_1-24/23.	Percent	0.063-0.350	4.5.6
Dielectric breakdown parallel to laminations, step-by-step test.	A	Kilovolts	60-100	4.5.2
	D-48/50	Kilovolts	40-100	4.5.2
Dielectric strength perpendicular to laminations, step-by-step test.	A	Volts per milligram	248-354 (1/8-inch thickness).	See ASTM D 229 except voltage applied as in 4.5.2.2 and electrodes 3/4-inch in diameter 1 inch long used when more than 50 kilovolts required for puncture. Sample size is 6 by 6 inches by thickness.
Dielectric constant at 1 megacycle, 1/8-inch thickness.	E-96/50		3.48-2.87	See 4.5.7
	D-48/50		3.91-3.19	See 4.5.7
Dissipation factor at 1 megacycle, 1/8-inch thickness.	E-96/50		0.019-0.010	See 4.5.7
	D-48/50		0.045-0.029	See 4.5.7
Insulation resistance	E-96/50	Megohms	350,000-775,000	See ASTM D 257 using tapered pin electrodes. Sample size is 2 by 3 inches by thickness.
	D_1-96/50	Megohms	435-250,000	See ASTM D 257 using tapered pin electrodes. Sample size is 2 by 3 inches by thickness.
	C-96/35/90	Megohms	105-32,800	See ASTM D 257 using tapered pin electrodes. Sample size is 2 by 3 inches by thickness.
Bonding strength	A	Pounds	1,000-2,260	See 4.5.5
	D-48/50	Pounds	1,020-2,170	See 4.5.5
Arc resistance	A	Seconds	Less than 15	Method 4011 of FED-STD-406

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